FLAT BASE CHANGE PRESERVES PERFECT MORPHISMS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17743-17746 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17748-17757, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE candidate: the whole proof is two exact-label citations plus the definitional split; flagged for the Chief alongside L16197/L16198 precedent.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemma-flat-base-change-pseudo-coherent; More on Algebra, lemma flat-base-change-finite-tor-dimension.

Source reference. Original lemma 0680: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17388–17390. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 066M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17757–17757. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A flat base change of a perfect morphism is perfect.*

Proof. This translates into the following algebra result: Let $A \to B$ be a perfect ring map. Let $A \to A'$ be flat. Then $A' \to B \otimes_A A'$ is perfect. This result for pseudo-coherent ring maps we have seen in Lemma 0680. The corresponding fact for finite tor dimension follows from More on Algebra, Lemma 066M. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.